Exact symmetry of the eight twist-derived kernels

On the fixed public p827 example, the certified degree17 endomorphism cycles through all48 nonzero twist seven-torsion points and all eight kernel lines. Native computation and independent PARI/GP agree. These are exact finite computations, not measured relation-solving costs.

Recorded quantity	Actual action	Identity control	Scalar2 control
Point order	48	1	3
Projective order	8	1	1
Direct point checks	49	—	—
Addition checks	2401	—	—
Direct kernel-root matches	8	—	—

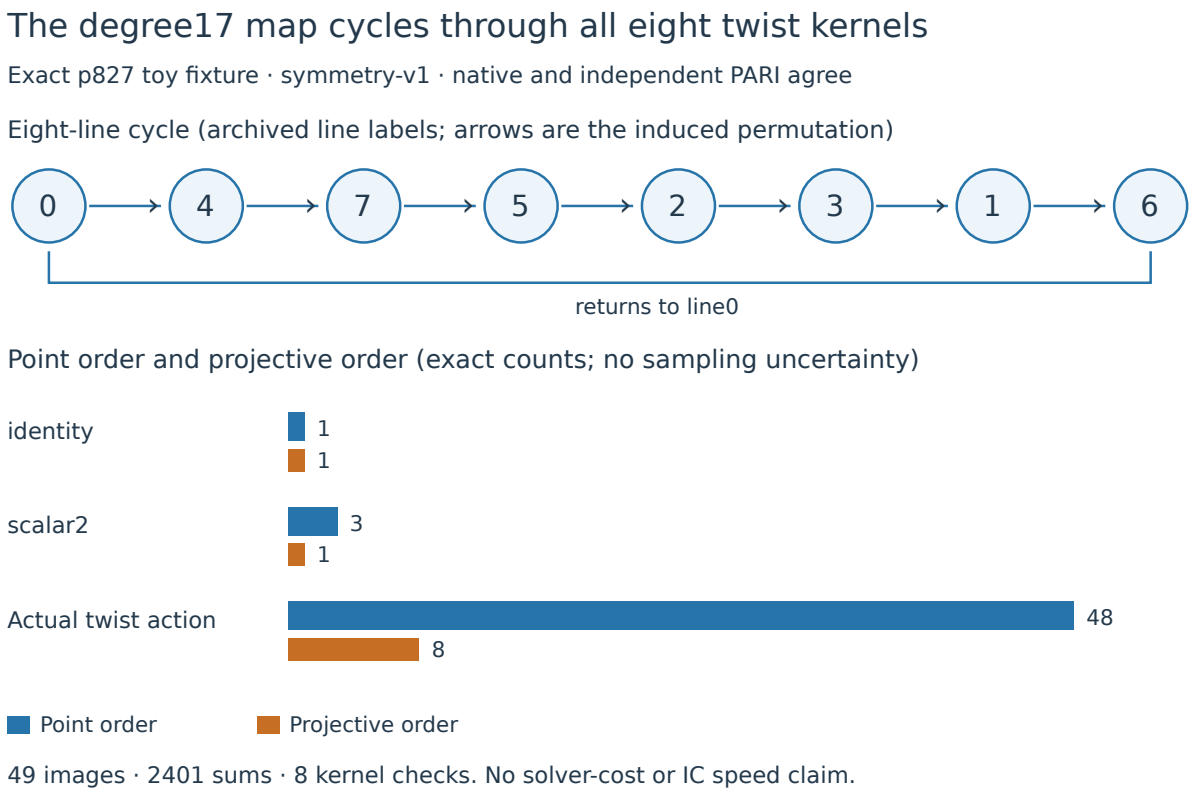


Figure 1: Recorded cycle and exact orders; run symmetry-v1. Complete finite populations, no statistical uncertainty.

The source is $E : y^2 = x^3 + x + 15$ over $\mathbb{F}_{827}$, ICV1 `icv1-fp10-t5-192cb216`, order823, generator(2,5). Its quadratic twist $T : Y^2 = X^3 + 4X + 120$ has order833. The unchanged archived basis of $T[7]$ is $T_1 = (0, 51)$ and $T_2 = (6, 130)$; labels(i, j) mean $iT_1 + jT_2$. The endomorphism is the previously certified $\alpha = (\pi + [1])/[7]$.

In that basis its independently recovered matrix and eighth power are

$$M = \begin{pmatrix} 3 & 1 \\ 5 & 5 \end{pmatrix} \pmod 7, \qquad M^2 - M + 3I = 0, \qquad M^8 = 3I.$$

The complete nonzero point set is one48-cycle. Kernel labels are $\langle T_1 + cT_2 \rangle$ for $c = 0, \ldots, 6$, followed by $\langle T_2 \rangle$ as line7. Their single cycle is $0 \to 4 \to 7 \to 5 \to 2 \to 3 \to 1 \to 6 \to 0$. Every image line was

checked by matching its three paired kernel x-coordinates against the archived v8 kernel polynomial roots.

Method, proof and scope

Native evaluation embeds into $E(\mathbb{F}_{827^2})$, applies the frozen normalized degree17 Vélu map and inverse-scale7 isomorphism, then unembeds. GP independently reconstructs Vélu with `ellisogeny` and evaluates the rational twist map $(X, Y) \mapsto (2R(X/2), YS(X/2))$. Each implementation reconstructs the matrix from point labels and checks every point, all2401 additions, complete point/line cycles, trace1 and determinant3. A corrupted coefficient is detected on42/49 points; swapping two line images yields2/8 mismatches.

The proof is finite: a cyclic basis (v, Mv) conjugates the matrix to $C = \left(\begin{smallmatrix} 0 & 4 \\ 1 & 1 \end{smallmatrix}\right)$ over$\mathbb{F}_7$. Exhaustion establishes $C^8 = 3I$, $C^{48} = I$, no shorter positive point period for any nonzero vector, and no shorter positive line period than8. The full raw orbit, point and kernel records are retained.

Four Lean declarations in `lean-v7/TwistKernelSymmetry.lean` prove these companion-module statements by kernel-checked finite arithmetic:

- `eighth_scalar`: eighth power is scalar3.
- `forty_eighth_identity`: forty-eighth power is the identity.
- `nonzero_point_period`: every nonzero point has period48.
- `projective_period`: every nonzero line has period8.

Pinned Lean4.34.1/mathlib revision `d13f23b723b8a846827a245b89c10fc7d3f11612`; frozen replay and fresh native transitive axiom audit pass with standard foundations only, no trusted `native_decide`. The concrete elliptic-curve/torsion-basis/map-to-matrix bridge remains unformalized. The v1 identity-certificate format fixes a different prime modulus and cannot certify this characteristic7 action; the exact native/GP and Lean evidence are the applicable certificates.

This supplies a finite symmetry relating all eight kernels. Induced explicit maps between their quotient curves and any saving in constructing or solving relations remain unmeasured. The point order48 on$T[7]$ differs from the known scalar action's order274 on$E(\mathbb{F}_{827})$. No degree1 automorphism, cheap rho fold, novelty, full-IC improvement or NIST applicability is inferred.

Reproducibility and execution

Source revision `ee305333dfb7c7acc19169794de12e6c6538a3be`; frozen protocol hash `49ac79e28023c2e4a23f27e299419b8cb15ac577ba50a531ad43d6e0a2ac07dd`. The frozen record was committed and Setec gate passed before data. Native and GP each completed their single300s-capped run on the managed Xeon8573C KVM host, one thread/CPU0/4GiB, L0; no clocks are admitted.

Flow: planned2, attempted2, completed2, not completed0, interrupted0, resumed0, excluded0. Both strict admissions pass. Unified export binds protocol, source, script, inputs and actual raw files with zero findings. Independent implementation shares author/host; no separate-runner review exists.

Raw archive is5432bytes/eight files, SHA256 `1f0992c77000041a0d7d2be6d8642b5d5e1614442b3d07abc4d7bd22c51e36d5`, in the user-confirmed `s3://setec-astronomy-artifacts/store`. See execution-v11 for member inventory, deterministic recipe, fresh remote verification and all exact commands. Synthetic native tests pass18/18; all standalone runner tests pass. The prior full-root-suite validation gap remains open separately.

Preserved development issues: missing finite-product Lean import was fixed; draft placeholders/short SHA were rejected before freeze; the initial export omitted host memory metadata and was retained, then corrected with no science changes. Fresh ECR readiness remains access-denied, so the documented authorized local fallback applies. No paid resources or permissions were changed.

Literature: Andrew Sutherland, MIT18.783 Fall2025 Lecture5 Theorems5.11/5.15, Lecture13 Theorems13.6/13.8, accessed/read2026-10-28; repository `RESEARCH_ISOGENY_LOOP_GLV.md` and atlas GLV/isogeny-loop lead. These support the standard construction and distinction from unknown-log recovery.

Canonical progress/IC/curve graphs were checked: this exact module census does not change a performance row, model identity or previously verified edge. Prior graphs and frozen waves remain unchanged.